Dynamics of Mechanical Systems Prof. Stefano Bruni - Academic Year 2022-2023
Programme

Note: Blue = all students; orange = group 1 only; green = group 2 only

Week	Lecturer	Day		Month	Time	Lect./Ex.	Group	Room	Title
1	Bruni	Monday	12	September	13.15-15.15	Lect.	1+2	B8 0.2	Course introduction. Equation of motion for a 1-degree of freedom (d.o.f.) system
1	Bruni	Tuesday	13	September	13.15-15.15	Lect.	1+2	B8 0.2	Non-linear equation of motion for a 1 d.o.f. system
1		Wednesday	14	September	10.15-12.15	CANC			
1	Bruni	Thursday	15	September	9.15-11.15	Ex.	2	L.03	Review of the basic kinematic relationships, equation of motion for a 1-d.o.f. system, Lagrange equation.
1	Zuin	Thursday	15	September	14.15-16.15	Ex.	1	LM.6	Review of the basic kinematic relationships, equation of motion for a 1-d.o.f. system, Lagrange equation.
1	Bruni	Friday	16	September	15.15-17.15	Lect.	1+2	L.15	Linearisation of the equation of motion for a 1-d.o.f. system. Lagrange equations for a multi-d.o.f. system.
2	Bruni	Monday	19	September	13.15-15.15	Lect.	1+2	B8 0.2	Equations of motion for multi-d.o.f. systems. Linearisation of the equations of motion
2	Bruni	Tuesday	20	September	15.15-17.15	Lect.	1+2	B8 0.2	Free and forced motion for a 1 d.o.f. system. The Frequency Response Function (FRF)
2	Zuin	Wednesday	21	September	10.15-12.15	Ex.	1	LM.6	Equations of motion for 1 d.o.f. systems and linearization
2	Zuin	Thursday	22	September	9.15-11.15	Ex.	1	L.03	Equations of motion for multi-d.o.f. systems and linearisation (1)
2	Bruni	Thursday	22	September	14.15-16.15	Ex.	2	LM.6	Equations of motion for 1 d.o.f. systems and linearization
2	Bruni	Friday	23	September	15.15-17.15	Lect.	1+2	L.15	Free motion of a multi-d.o.f. system
3	Bruni	Monday	26	September	13.15-15.15	Lect.	1+2	B8 0.2	Forced motion for a multi-d.o.f. system. The FRF matrix. Periodic forcing
3	Bruni	Tuesday	27	September	15.15-17.15	Lect.	1+2	B8 0.2	The modal approach for a lumped parameter system. Resonances and anti-resonances.
3	Zuin	Wednesday	28	September	10.15-12.15	CANC			*Canceled due to BSc graduation exams*
3	Zuin	Thursday	29	September	9.15-11.15	Ex.	1	L.03	Equations of motion for a n-d.o.f. system with constraint displacement and linearisation (2)
3	Bruni	Thursday	29	September	14.15-16.15	Ex.	2	LM.6	Equations of motion for multi-d.o.f. systems and linearisation (1)
3	Bruni	Friday	30	September	15.15-17.15	Ex.	2	L.15	Equations of motion for a n-d.o.f. system with constraint displacement and linearisation (2)
4	Bruni	Monday	3	October	13.15-15.15	Lect.	1+2	B8 0.2	Modal identification techniques. Part 1: introduction and and experimental estimate of the system's FRF
4	Bruni	Tuesday	4	October	15.15-17.15	Lect.	1+2	B8 0.2	Modal identification techniques. Part 2: identification of modal parameters.
4	Zuin	Wednesday	5	October	10.15-12.15	Ex.	1	LM.6	Equations of motion for multi-d.o.f. systems and linearisation (3)
4		Thursday	6	October	9.15-11.15	CANC			*Canceled due to MSc graduation exams*
4		Thursday	6	October	14.15-16.15	CANC			*Canceled due to MSc graduation exams*
4	Bruni	Friday	7	October	15.15-17.15	Ex.	2	L.15	Equations of motion for multi-d.o.f. systems and linearisation (3)
5	Zuin	Monday	10	October	13.15-15.15	Lab.	1	B8 0.2	Computer-aided solution for the equations of motion of a 2-n d.o.f. system (1)
5	Bruni	Tuesday	11	October	15.15-17.15	Lab.	2	B8 0.2	Computer-aided solution for the equations of motion of a 2-n d.o.f. system (1)
5	Zuin	Wednesday	12	October	10.15-12.15	Ex.	1	LM.6	Equations of motion for multi-d.o.f. systems and linearisation (4)
5	Zuin	Thursday	13	October	9.15-11.15	Lab.	1	L.03	Computer-aided solution for the equations of motion of a multi-d.o.f. system
5	Bruni	Thursday	13	October	14.15-16.15	Ex.	2	LM.6	Equations of motion for multi-d.o.f. systems and linearisation (4)
5	Bruni	Friday	14	October	15.15-17.15	Lab.	2	L.15	Computer-aided solution for the equations of motion of a multi-d.o.f. system
6	Bruni	Monday	17	October	13.15-15.15	Lect.	1+2	B8 0.2	Distributed parameter systems. Rigid vs. deformable bodies. Strain and stress components.
6	Bruni	Tuesday	18	October	15.15-17.15	Lect.	1+2	B8 0.2	Distributed parameter systems. Mechanics of linear beams. Internal forces in beams.
6	Zuin	Wednesday	19	October	10.15-12.15	Ex.	1	LM.6	Calculation of constraint forces; calculation of spring pre-loads in the equilibrium position
6	Zuin	Thursday	20	October	9.15-11.15	Ex.	1	L.03	Solved problems in preparation of the written exam
6	Bruni	Thursday	20	October	14.15-16.15	Ex.	2	LM.6	Calculation of constraint forces; calculation of spring pre-loads in the equilibrium position
6	Bruni	Friday	21	October	15.15-17.15	Ex.	2	L.15	Solved problems in preparation of the written exam
7	Bruni	Monday	24	October	13.15-15.15	Lect.	1+2	B8 0.2	Axial vibration of beams
7	Bruni	Tuesday	25	October	15.15-17.15	Lect.	1+2	B8 0.2	Theory of bending in slender beams (Euler-Bernoulli beams)
7	Bruni	Wednesday	26	October	10.15-12.15	Lect.	1+2	LM.6	Bending vibration of slender beams
7	Zuin	Thursday	27	October	9.15-11.15	Lab.	1	L.03	Solved problems in preparation of the written exam (2)

7	Bruni	Thursday	27	October	14.15-16.15	Lab.	2	LM.6	Solved problems in preparation of the written exam (2)
7	Bruni	Friday	28	October	15.15-17.15	Ex.	1+2	L.15	Solved problems in preparation of the written exam (3)
8		Monday	31	October	13.15-15.15	CANC			*Canceled (holiday)*
8		Tuesday	1	November	15.15-17.15	CANC			*Canceled (holiday)*
8	Zuin	Wednesday	2	November	10.15-12.15	Ex.	1+2	1+2	Solved problems in preparation of the written exam (4)
8	Zuin	Thursday	3	November	9.15-11.15	Ex.	1	L.03	Vibrations of distributed-parameter systems
8	Bruni	Thursday	3	November	14.15-16.15	Ex.	2	LM.6	Vibrations of distributed-parameter systems
8		**Friday**	**4**	**November**	**8.15-11.15**	**Test**	**1+2**	**t.b.d.**	**In-course test (problem 1)**
8	Zuin	Wednesday	9	November	10.15-12.15	Ex.	1	LM.6	Vibrations of distributed-parameter systems
9	Bruni	Thursday	10	November	9.15-11.15	Lect.	1+2	L.03	Theory of torsion in beams. Torsional vibrations.
9	Bruni	Thursday	10	November	14.15-16.15	Ex.	2	LM.6	Vibrations of distributed-parameter systems
9	Bruni	Friday	11	November	15.15-17.15	Lect.	1+2	L.15	The modal approach for distributed parameter systems (beams)
10	Bruni	Monday	14	November	13.15-15.15	Lect.	1+2	B8 0.2	The finite element method. Introduction. The Euler-Bernoulli beam element. Shape functions.
10	Bruni	Tuesday	15	November	15.15-17.15	Lect.	1+2	B8 0.2	The finite element method. Mass and stiffness matrices for the Euler-Bernoulli beam element.
10	Bruni	Wednesday	16	November	10.15-12.15	CANC			
10	Zuin	Thursday	17	November	9.15-11.15	Ex.	1	L.03	Vibrations of distributed-parameter systems
10	Bruni	Thursday	17	November	14.15-16.15	Ex.	2	LM.6	Vibrations of distributed-parameter systems
10	Bruni	Friday	18	November	15.15-17.15	Lect.	1+2	LM.6	The finite element method. Assembling of mass and stiffness matrices. Structural damping. Coordinate partitioning.
11	Bruni	Monday	21	November	13.15-15.15	Lect.	1+2	B8 0.2	Solid finite elements: shape functions, mass and stiffness matrices, distortion.
11	Bruni	Tuesday	22	November	15.15-17.15	Ex.	1+2	L.15	Vibration of distributed-parameter systems
11		Wednesday	23	November	10.15-12.15	CANC			
11	Zuin	Thursday	24	November	9.15-11.15	Lab.	1+2	L.03	Vibration of distributed-parameter systems
11	Zuin	Thursday	24	November	14.15-16.15	Lab.	1+2	LM.6	Analysis of vibrating beam systems using the Finite Element method.
11	Bruni	Friday	25	November	15.15-17.15	Lect.	1+2	L.15	Spatial Kinematics and dynamics of a rigid body (1)
12	Bruni	Monday	28	November	13.15-15.15	Lect.	1+2	B8 0.2	Spatial Kinematics and dynamics of a rigid body (2)
12	Bruni+Zuin	Tuesday	29	November	15.15-17.15	Lab.	1+2	B8 0.2	Introduction to the year work
12		Wednesday	30	November	10.15-12.15	CANC			*Opening of Academic year*
12	Bruni	Thursday	1	December	9.15-11.15	Lect.	1+2	L.03	Stability of mechanical systems in force fields. 1-d.o.f. systems
12	Bruni+Zuin	Thursday	1	December	14.15-16.15	Lab.	1+2	LM.6	Preparation of the year work (1)
12	Bruni	Friday	2	December	15.15-17.15	Lect.	1+2	L.15	Stability of mechanical systems in force fields. 2-n d.o.f. systems
13	Bruni+Zuin	Monday	5	December	13.15-15.15	Lab.	1+2	B8 0.2	Preparation of the year work (2) Periodic excitation - Fourier series and Fourier transform
13	Bruni	Tuesday	6	December	15.15-17.15	Lect.	1+2	B8 0.2	Rolling contact between elastic bodies. Tyre-road forces. Wheel-rail contact forces.
13	Bruni	Monday	12	December	15.15-17.15	Lect.	1+2	B8 0.2	The bicycle model for a car in a curve.Stability analysis.
13	Bruni	Tuesday	13	December	15.15-17.15	Lect.	1+2	B8 0.2	Stability of a railway wheelset.
13	Bruni	Wednesday	14	December	10.15-12.15	Lect.	1+3	LM.6	Stability of a solid body exposed to wind action
13	Bruni+Zuin	Thursday	15	December	9.15-11.15	Lab.	1+2	L.03	Preparation of the year work (3)
13	Zuin	Thursday	15	December	14.15-16.15	Lab.	1+2	LM.6	Modal identification techniques
13	Bruni	Friday	16	December	15.15-17.15	Ex.	1+2	L.15	Recap for written exam - Problem n.1
14	Bruni+Zuin	Monday	19	December	13.15-15.15	Lab.	1+2	B8 0.2	Preparation of the year work (4)
15		Tuesday	20	December	15.15-17.15	CANC			*Canceled due to MSc graduation exams*
15	Zuin	Wednesday	21	December	10.15-12.15	Ex.	1+2	LM.6	Recap for written exam - Problem n.2
15		Thursday	22	December	9.15-11.15	CANC			
15		Thursday	22	December	14.15-16.15	CANC			